



5.1 INTRODUCTION

In earlier grades you worked with triangles and quadrilaterals — figures built from straight edges, where almost any claim could be settled by measuring the sides and adding up the angles.

This chapter takes the next step, into the circle. It has no straight edges, no corners, and no pieces to count. Yet nearly everything you are about to prove about it follows from one sentence saying what a circle is. Before that sentence, something to explain.

People have been drawing circles for as long as they have been drawing anything. Among the rock paintings at Gudāhāndi in Odisha, beside the triangles and the squares, there are circles and ovals. The shape was there to be copied: a raindrop spreads across still water as a circle, the cut face of a stem is a circle, and so, to the eye, are the full moon and the sun.

EXAMPLE 1

Jamuna cuts a disc out of paper. Nothing on it marks the centre.

She folds the disc so that the boundary falls exactly onto itself, and presses the crease flat. She opens it out, folds it again a different way, and presses again.

The two creases cross at a single point — and that point is the centre.

Cut a fresh disc, carelessly, and it happens again. Fold a third time and the third crease runs through the same point. Folding can go on confirming this for as long as you have paper, but it can never tell you *why* the creases meet at all, nor why they meet at the centre rather than somewhere near it. For that you need to say what a circle is — and by the end of Section 5.3 you will settle it in three lines.



A definition is where a proof starts: say exactly what a shape is, and its properties stop being coincidences.



THINK AND REFLECT

Jamuna's second fold was *a different way*. Does it matter how different?

Try a pair of folds that are nearly the same as each other, and a pair that are nearly at right angles. Both pairs cross at the centre. Which pair locates it more surely on paper, and why?



5.2 WHAT A CIRCLE IS

When we speak of a circle, a triangle or a square, we take the figure to be drawn on a flat sheet — a **plane**. Everything in this chapter happens on one.

Fix a point of that plane, and fix a distance. Now collect every point of the plane that lies exactly that far from the fixed point. That collection is a circle, and it is the whole of the definition.

CIRCLE

A **circle** is the set of all points of a plane that are at a fixed distance from a fixed point of that plane. The fixed point is the **centre**; the fixed distance is the **radius**.

Notice what the definition does *not* say. It says nothing about roundness, nothing about smoothness, and nothing about how to draw one. It fixes a point and a distance, and everything else in this chapter — every symmetry, every theorem — has to be squeezed out of those two things.

A set of points singled out by a condition like this is called the **locus** of points satisfying the condition. In that language, a circle is the locus of points at a given distance from a given point. The word will earn its keep in Section 5.4, where the condition changes.

The parts of a circle have names. Take two points A and B on it; the segment joining them is a **chord**. Join the two ends of a chord to the centre and an angle appears there: the chord AB is said to **subtend** the angle $\angle AOB$ at the centre. A

chord that happens to pass through the centre is a **diameter**. Fig. 5.1 shows all three.

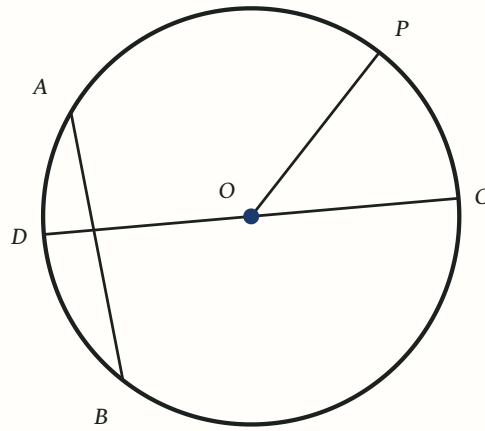


Fig. 5.1 A circle with centre O . Here OP is a radius, AB is a chord, and CD is a chord through the centre — a diameter.

EXAMPLE 2

A circle has centre O and radius 10 cm. Three points are marked, with $OP = 10$ cm, $OQ = 7$ cm and $OR = 13$ cm. Which of them lie on the circle?

The definition puts one question to a point, and only one: is its distance from the centre equal to the radius? So set each distance beside $r = 10$ cm.

P	$OP = 10$, and $r = 10$	on the circle
Q	$OQ = 7$, which is $< r$	inside
R	$OR = 13$, which is $> r$	outside

Only P lies on the circle. Notice that being inside or outside is not a near miss that might be argued about: the definition allows no degrees. A point either stands at distance r from the centre or it is no part of the circle at all.

A diameter is a chord of a particular kind, and its length follows from the definition alone: both ends lie at distance r from the centre, and the centre lies between them. Every diameter of a given circle is $2r$ long.

No chord is longer. A chord that misses the centre still has both ends at distance r from it, so going from one end to the other by way of the centre measures $2r$ — and going straight across is shorter. Nothing here was measured: the argument used the definition, and that a straight path is the shortest one.



THINK AND REFLECT

1. What is the length of the longest chord of a circle of radius 5 units?
2. Is there a shortest chord? Try to name one, and see what goes wrong.



The second question is not the first one turned around. Shrink a chord towards a point of the circle and ask, at each stage, whether what you still have is a chord.

5.3 SYMMETRIES OF A CIRCLE

A circle looks the same from every direction, and that is not a vague compliment — it is a statement with consequences, and it follows from the definition in a line. There are two kinds of sameness to separate.

Watch a wheel go past. At one moment some point of the rim is touching the road; a moment later some point of the rim is touching the road again. Is it the same point? Nothing you can see decides it. Chalk a mark on the rim and you can follow it, as in Fig. 5.2 — but rub the mark out and the two pictures become one picture.

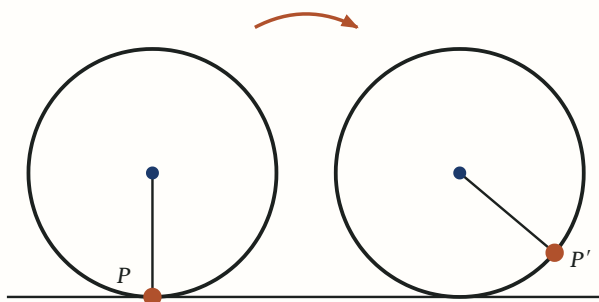


Fig. 5.2 The same wheel, a moment later. Only the chalk mark records that anything moved.

Turn a circle about its centre, through any angle you like, and you get the circle back — not another circle of the same size, but the very same set of points. The reason is the definition and nothing else. Turning about O leaves every point's distance from O unchanged, so a point at distance r goes to a point at distance r ; the set of all points at distance r is carried onto itself.

No other shape in this book does that. Turn a square about its centre and it comes back to itself four times in a full turn, at multiples of 90° , and at no other angle. A circle comes back at *every* angle. We say it has **complete rotational symmetry**.

Now the other kind of sameness. Draw a circle, cut it out, and fold it so that the boundary falls exactly onto itself. Open it out: the crease is a line of **reflection symmetry**. It is Jamuna's fold from the first page of this chapter, and we are at last in a position to say what it does.

BACK TO THE FOLDED PAPER

Call the crease ℓ . Folding along ℓ carries the circle onto itself, and folding does not change distances. So the point where the centre lands — call it O' — is again a point from which every point of the circle is at distance r . Could O' be anywhere other than O ? If it were, every point of the circle would be equidistant from two different points, and so would lie on the perpendicular bisector of OO' — the whole circle would sit on one straight line. It does not. So $O' = O$. And the only points a fold leaves where they were are the points of the crease itself.

So O lies on ℓ . Every crease of this kind is a line through the centre — a diameter. Two different creases are two different lines through O , and two different lines meet at no more than one point. They meet at O . That is why Jamuna's construction cannot fail, and why her third fold was never going to tell her anything.

LINES OF SYMMETRY OF A CIRCLE

A line is a line of reflection symmetry of a circle exactly when it passes through the centre. The lines of symmetry of a circle are its diameters — and there are infinitely many of them.

EXERCISE SET 5.1

1. A circle has radius 6.5 cm.
 - (i) How long is its longest chord?
 - (ii) Can a chord of it be 14 cm long? Give a reason.
2. Say whether each statement is true, and give a reason in one line.
 - (i) Every diameter is a chord.
 - (ii) Every chord is a diameter.
 - (iii) A circle has two lines of symmetry.
 - (iv) A circle has exactly one centre.
3. Through what angles can a square be turned onto itself? A regular pentagon? A regular hexagon? How many lines of symmetry has each?
4. Jamuna folds her disc a third time. Must the third crease pass through the point where the first two crossed? Say why.

5.4 HOW MANY CIRCLES?

In Section 5.2 the word **locus** was introduced and then left alone: a circle is the locus of points at a given distance from a given point. Change the condition and you get a different locus. Here is the change worth making.

Instead of one fixed point, take two. Let A and B be points of the plane, and ask a question not about points but about circles: how many circles pass through both of them?

At least one does. If a circle passes through A and B , its centre O must satisfy $OA = OB$ — and one point obviously does, the midpoint of AB . Take that as the centre, with radius half of AB , and the circle drawn passes through both. For that circle, AB is a diameter.

Are there others? Notice that the question has quietly turned back into a question about points: which points of the plane are equidistant from A and B ?

You met the answer in an earlier grade. They are the points of the **perpendicular bisector** of AB — and there are infinitely many of them.

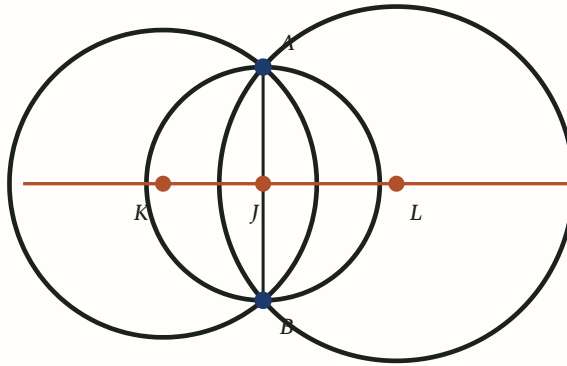


Fig. 5.3 Three of the circles through A and B . Every centre lies on the perpendicular bisector of AB , and every point of that line is the centre of one.

That answer needs stating with some care, because it is really two statements and a locus needs both of them:

Every point of the perpendicular bisector is equidistant from A and B ; and every point equidistant from A and B lies on the perpendicular bisector. The first says that no point of the line is missed out. The second says that no point outside it sneaks in. Drop either half and "the locus of" becomes the wrong phrase — you would have described some of the centres, or too many.

With both halves in hand the count is settled. The centres of the circles through A and B are exactly the points of the perpendicular bisector of AB : one circle for each point of a line, and so infinitely many.

CIRCLES THROUGH TWO POINTS

Infinitely many circles pass through any two given points. Their centres are precisely the points of the perpendicular bisector of the segment joining them, and the smallest of those circles has that segment as a diameter.



THINK AND REFLECT

1. As the centre travels along the bisector away from AB , do the radii grow or shrink? Is there a largest circle through A and B ?
2. Which is the smallest, and what is its radius?
3. Far out along the bisector, the circle still passes through A and B . Does the arc between them look more curved or less?

For the third, ask what the arc is turning into. Nothing in the argument ever lets it become a straight line — but see how close it is willing to get.

Two points, then, are not enough to pin a circle down. Three might be — and the answer turns out to depend on how the three are arranged.

The first thing to notice about three points is that they need not have any circle through them at all.

Suppose A , B and C lie on one straight line — that they are **collinear**. A circle through all three would need a centre equidistant from A and B , so lying on the perpendicular bisector of AB ; and equidistant from B and C , so lying on the perpendicular bisector of BC . But both of those bisectors are perpendicular to the same line, which makes them parallel, and they are not the same line, because AB and BC have different midpoints. Parallel and distinct: they never meet. There is no such centre, and so no such circle.

Three points in a row, then, are hopeless. Rule that case out and everything changes.

THEOREM 1

Through any three points that are not collinear there passes exactly one circle.

Why is this true? The argument is the one just used, run forwards instead of backwards. Suppose some circle passes through A , B and C , and call its centre O .

Since $OA = OB$, the point O lies on the perpendicular bisector of AB .

Since $OA = OC$, the point O lies on the perpendicular bisector of AC .

Now A , B and C are not collinear, so AB and AC point in different directions, and the two bisectors are therefore not parallel. Two non-parallel lines of a plane meet at exactly one point. So there is only one candidate for O — which settles the *exactly one* before we have even shown that a circle exists.

And it does exist. Take that meeting point as centre and draw the circle of radius OA . Because O is on the bisector of AB we have $OB = OA$, and because it is on the bisector of AC we have $OC = OA$. So B and C lie on that circle as well, and all three points are on it.



THINK AND REFLECT

Draw three collinear points A , B , C and construct the perpendicular bisectors of AB and BC . Check that they really are parallel.

Now nudge C off the line, a little at a time. The two bisectors begin to meet. Where does their meeting point go as C approaches the line again — and what is happening to the circle?

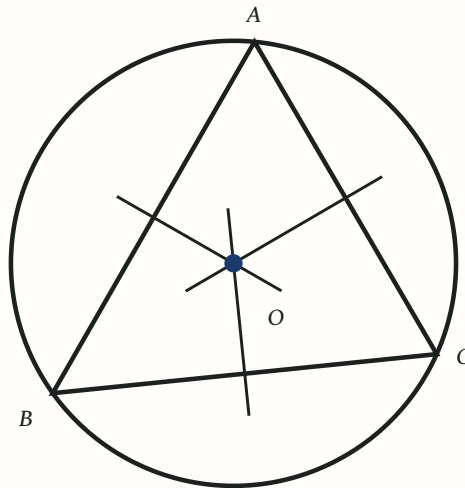


Fig. 5.4 The circumcircle of triangle ABC . The perpendicular bisectors of the three sides meet at the centre.

Three non-collinear points also determine a triangle, and what we have just built is the circle through its vertices. Both the circle and its centre have names.

The circle through the vertices of a triangle is its **circumcircle**, and its centre is the **circumcentre** — the point where the perpendicular bisectors of the sides meet. The circle is said to **circumscribe** the triangle, and the triangle to be **inscribed** in the circle.

Note what the argument gave us along the way. All three perpendicular bisectors pass through O , though only two were used to find it; the third has no choice in the matter, since O is equidistant from B and C too.

Theorem 1 says the circumcircle exists and that there is only one. It says nothing about *where* the centre lands — and the answer is not the one you might guess, because the circumcentre need not lie inside the triangle at all.

EXAMPLE 3

A triangle has vertices $A(0,0)$, $B(6,0)$ and $C(0,8)$. Find its circumcentre, and the radius of its circumcircle.

The circumcentre is where the perpendicular bisectors meet, so two of them are enough. Which two? The ones that are least work — and here two sides lie along the axes, so their bisectors can be written down without any calculation.

bisector of AB AB runs along the x -axis, 0 to 6 $x = 3$

bisector of AC AC runs along the y -axis, 0 to 8 $y = 4$

Those two lines cross at $O(3,4)$, so that is the circumcentre. The radius is its distance from any one vertex:

$$OA = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

Check the other two rather than assume them: OB and OC come to 5 as well. All three agree, as Theorem 1 says they must.

EXAMPLE 4

Is there a circle through $P(1,2)$, $Q(3,6)$ and $R(5,10)$?

Theorem 1 promises one for any three points that are not collinear, so the real question is whether these three are in a row. Compare the step from P to Q with the step from Q to R .

P to Q across 2, up 4 2 up per 1 across

Q to R across 2, up 4 2 up per 1 across

The two steps rise at the same rate and share the point Q , so all three points lie on one straight line. There is no circle through them — and the reason is the one given above: the perpendicular bisectors of PQ and QR are parallel, so they never meet to give a centre.

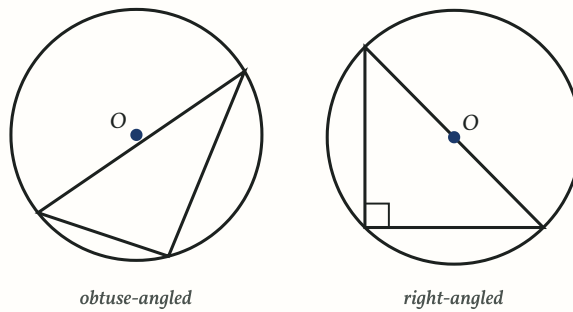


Fig. 5.5 Where the circumcentre lands. Outside an obtuse-angled triangle; on the midpoint of the hypotenuse of a right-angled one.

For an acute-angled triangle the circumcentre lies inside — Fig. 5.4 is one such. For an obtuse-angled triangle it lies outside the triangle altogether. And for a right-angled triangle it lands exactly on the midpoint of the hypotenuse, which is another way of saying that the hypotenuse is a diameter of the circumcircle.



That last case is not a coincidence, and it is not proved here. Section 5.8 will hand it to you in a line.

EXERCISE SET 5.2

1. Draw $\triangle ABC$ with $AB = 5$ cm, $\angle A = 70^\circ$ and $\angle B = 60^\circ$. Draw its circumcircle. Is the centre inside the triangle or outside it?
2. Draw $\triangle ABC$ with $AB = 5$ cm, $\angle A = 100^\circ$ and $AC = 4$ cm. Draw its circumcircle. Inside or outside this time, and does that agree with Fig. 5.5?

3. Draw $\triangle ABC$ with $AB = 6$ cm, $BC = 7$ cm and $CA = 7$ cm, and let O be its circumcentre. Measure OA , OB and OC . What must you find, and why?
4. What is the least possible radius of a circle passing through two points 8 cm apart?
5. Can a straight line cut a circle in three distinct points? Give a reason.
- *6. The circumcircle of a triangle ABC is drawn. Can a different triangle, congruent to ABC , have the same circumcircle? How many such triangles are there?

5.5

CHORDS AND THE ANGLES THEY SUBTEND

Tie a thread between two points of a wheel and pull it tight. What you have is a chord. Join its two ends to the centre, and the thread subtends an angle there — the angle named in Section 5.2.

Now turn the wheel. The thread goes round with it and comes to rest across a different pair of points; but it is the same thread, so its length has not changed. Has the angle at the centre changed with it?

Section 5.3 all but answers this. Turning the circle about its centre carries the circle onto itself and carries the thread along without stretching it, so nothing in the picture is different except which part of the rim the ends are tied to. The angle cannot have moved.

That is a good reason to believe what follows. It is not yet a proof of it, because the argument assumed the second chord was the first one turned — and a chord of the same length lying somewhere else on the circle has not been shown to be that. Congruent triangles settle the matter without the assumption.

THEOREM 2

Equal chords of a circle subtend equal angles at the centre.

Let the chords be AB and DE , with $AB = DE$, in a circle of centre C and radius r . We want $\angle ACB = \angle DCE$. Join all four ends to the centre, as in Fig. 5.6, and compare the two triangles that appear.

$CA = CB = r$ and $CD = CE = r$, so $CA = CD$ and $CB = CE$.

$AB = DE$ is given.

So $\triangle CAB \cong \triangle CDE$, by **SSS**.

Corresponding angles of congruent triangles are equal, and $\angle ACB$ and $\angle DCE$ are corresponding. So the two chords subtend the same angle at the centre.

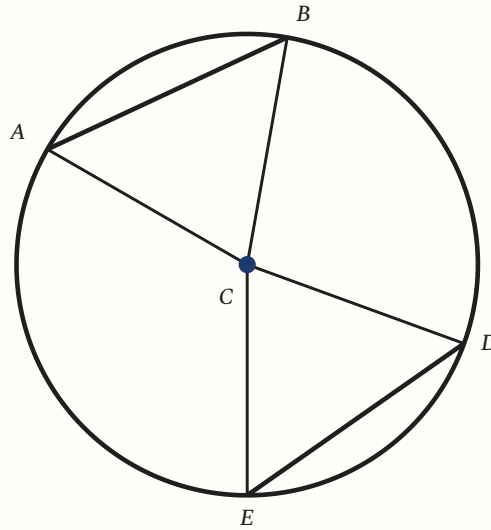


Fig. 5.6 Two chords of the same length. The triangles CAB and CDE have three pairs of equal sides.

Now run the argument backwards. Suppose instead that the two angles at the centre are equal — must the chords be?

THEOREM 3

Chords of a circle that subtend equal angles at the centre are equal in length.

$CA = CD = r$ and $CB = CE = r$.

$\angle ACB = \angle DCE$ is now what is given, and it is the angle *between* those two pairs of sides.

So $\triangle CAB \cong \triangle CDE$, by **SAS**, and therefore $AB = DE$.

Notice how little separates the two proofs: the same figure, the same two triangles, the same four radii. All that changes is which fact is given and which is wanted — and with it, the congruence rule that does the work.

EXAMPLE 5

A chord of a circle of radius 9 cm subtends 60° at the centre. How long is the chord? And if a second chord of the same circle is also 9 cm long, what angle does it subtend?

Join the ends of the chord to the centre. The two radii and the chord make a triangle, and here it is better than isosceles.

two sides both are radii, so 9 cm each **isosceles**

apex angle 60° , at the centre **equilateral**

The base angles of an isosceles triangle are equal, and if the apex takes 60° the remaining 120° splits into 60° each. All three angles equal makes all three sides equal, so the chord is 9 cm — exactly the radius.

The second part needs no calculation. The other chord is the same length, so by Theorem 2 it subtends the same angle.

EXERCISE SET 5.3

1. Show that the triangle formed by a chord of a circle and the two radii to its ends is isosceles.
2. Two such triangles, in the same circle, have bases of equal length. Show that they are congruent.
3. Two chords of a circle of radius 10 cm each subtend 60° at the centre. How long is each chord? (No calculation should be needed.)
4. A chord of a circle of radius 8 cm subtends a right angle at the centre. Find its length.
- *5. One chord subtends 60° at the centre and another subtends 120° . Is the second chord twice as long as the first? Give a reason.

5.6

CHORDS, MIDPOINTS AND PERPENDICULARS

Section 5.5 took the triangle made by a chord and the two radii to its ends, and got two theorems out of it. That triangle has more to give. It is isosceles — which is what Exercise Set 5.3 asked you to show — and an isosceles triangle is symmetric about the line through its apex and the midpoint of its base.

Here the apex is the centre. So draw a circle with centre C , take any chord AB , and let M be the midpoint of AB . What can be said about the segment CM ?

THEOREM 4

The segment joining the centre of a circle to the midpoint of a chord is perpendicular to that chord.

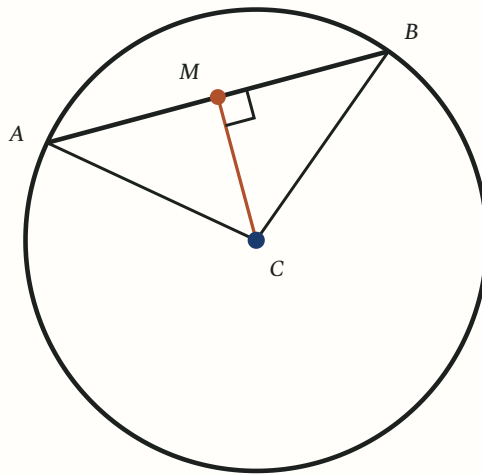


Fig. 5.7 M is the midpoint of the chord AB . The segment CM meets the chord at a right angle.

Why is this true? Compare the two triangles that CM cuts the big one into.

$CA = CB$, both being radii.

$AM = BM$, because M is the midpoint.

CM is common to both.

So $\triangle CMA \cong \triangle CMB$, by **SSS**.

Congruent triangles give $\angle CMA = \angle CMB$. But those two angles sit side by side on the straight line AB , so they add to 180° . Two equal angles adding to 180° are 90° each. So CM is perpendicular to AB .

Now the converse. Drop a perpendicular from the centre onto a chord instead, and ask where it lands.

THEOREM 5

The perpendicular from the centre of a circle to a chord bisects the chord.

$CA = CB$, both radii — and in the right-angled triangles CMA and CMB these are the hypotenuses.

CM is common, and $\angle CMA = \angle CMB = 90^\circ$ is now what is given.

So $\triangle CMA \cong \triangle CMB$, by **RHS**, and therefore $AM = BM$.

Three things have been in play: that the segment runs from the centre, that it meets the chord at a right angle, and that it hits the midpoint. Theorem 4 takes the first and third and delivers the second; Theorem 5 takes the first and second and delivers the third. Any two of the three force the remaining one — and the last pairing is the first question below.

EXAMPLE 6

A chord of length 16 cm is drawn in a circle of radius 10 cm. How far is the chord from the centre?

Distance from the centre means the perpendicular distance, and Theorem 5 says that perpendicular lands on the midpoint of the chord. That is what makes the calculation possible: it produces a right-angled triangle with a radius as its hypotenuse and half the chord as one leg.

half the chord $\frac{1}{2} \times 16$ 8 cm

hypotenuse a radius of the circle 10 cm

The Baudhāyana–Pythagoras theorem [set out in the *Śulbasūtra*, centuries before Pythagoras] now gives the remaining side:

$$d^2 = 10^2 - 8^2 = 100 - 64 = 36, \quad d = 6$$

The chord lies 6 cm from the centre — less than the radius, as it must be: a line 10 cm out would only touch the circle.

EXERCISE SET 5.4

1. Explain why the perpendicular bisector of a chord must pass through the centre of the circle. (This is the third of the three pairings.)
2. An isosceles triangle ABC with $AB = AC$ is inscribed in a circle. Show that the altitude from A to BC passes through the centre.
3. A chord of length 24 cm lies 5 cm from the centre of a circle. Find the radius.
4. Two parallel chords, of lengths 6 cm and 8 cm, lie on opposite sides of the centre of a circle of radius 5 cm. Find the distance between their midpoints.
- *5. In the previous question, what would the distance be if the two chords lay on the *same* side of the centre?

5.7 HOW FAR A CHORD LIES FROM THE CENTRE

The distance from a point to a line means the perpendicular distance — the shortest way across. So the distance from the centre to a chord is measured along the perpendicular from the centre, and Theorem 5 has already told us where that perpendicular lands: on the midpoint of the chord.

Take a paper circle and fold a piece of the boundary inwards; the crease is a chord. Fold again so that the two ends of that crease meet, and open both folds. The second crease crosses the first at its midpoint, square on. Theorems 4 and 5, in paper.

Now make a second chord of the same length somewhere else on the circle, and measure how far it lies from the centre. The two distances come out equal. Try another pair and they come out equal again.

Is that convincing? It is the same trap as Jamuna's folded disc on the first page of this chapter. A statement can hold in every case you have patience to try and still be false; and even when it is true, a stack of measurements never says *why*. So here is why.

THEOREM 6

Chords of a circle that have the same length lie at the same distance from the centre.

Let AB and FG be chords of a circle with centre C and radius r , with $AB = FG$. Let E and H be their midpoints, so that by Theorem 4 the segments CE and CH are perpendicular to the chords — and are therefore the two distances we are comparing.

$AE = \frac{1}{2}AB = \frac{1}{2}FG = FH$, since E and H are the midpoints.

$CA = CF = r$, and these are the hypotenuses.

$\angle CEA = \angle CHF = 90^\circ$.

So $\triangle CEA \cong \triangle CHF$, by **RHS**, and therefore $CE = CH$.

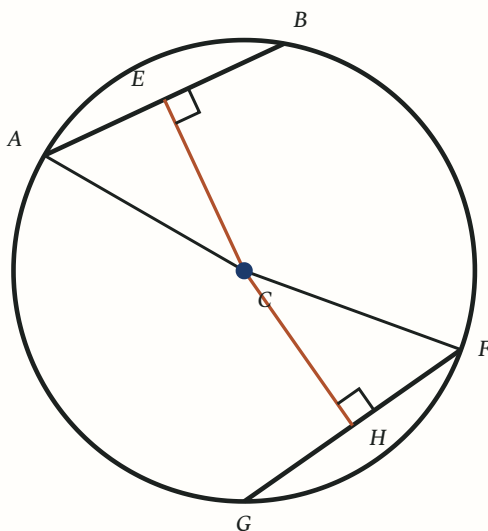


Fig. 5.8 Equal chords AB and FG . The perpendiculars CE and CH have the same length.

There is a second route. Triangles CAB and CFG are congruent by **SSS** — two pairs of radii and the given equal chords — and CE and CH are corresponding altitudes of congruent triangles. Either argument will do; the first has the advantage of never leaving the right-angled halves.

Turn it round and the converse follows from the same two triangles, with the given fact and the wanted one exchanged once more.

THEOREM 7

Chords of a circle that lie at the same distance from the centre are equal in length.

$CE = CH$ is now given, and $CA = CF = r$.

$\angle CEA = \angle CHF = 90^\circ$, so **RHS** gives $\triangle CEA \cong \triangle CHF$ again.

Hence $AE = FH$, and doubling gives $AB = FG$.

Both theorems come out of one relation, which is worth having on its own. Look again at the right-angled triangle CEA in Fig. 5.8. Its hypotenuse is the radius r , one leg is the distance $d = CE$, and the other leg is half the chord. The Baudhāyana–Pythagoras theorem gives

$$r^2 = d^2 + \left(\frac{1}{2} AB\right)^2, \quad \text{so} \quad AB = 2\sqrt{r^2 - d^2}$$

Read that formula and both theorems fall out of it at once. In a given circle r is fixed, so the length of a chord depends on d and on nothing else whatever. Equal distances therefore force equal lengths, and equal lengths force equal distances. Theorems 6 and 7 are the two directions of a single statement.

EXAMPLE 7

A chord lies 8 cm from the centre of a circle of radius 17 cm. How long is it?

This is the relation just proved, used in the direction it was built for. Put $r = 17$ and $d = 8$ into it and there is nothing left to decide.

$$\begin{aligned}\text{chord} &= 2\sqrt{r^2 - d^2} = 2\sqrt{17^2 - 8^2} \\ &= 2\sqrt{289 - 64} = 2\sqrt{225} = 2 \times 15 = 30\end{aligned}$$

The chord is 30 cm long. Compare it with the diameter, 34 cm: it is shorter, as every chord must be, and it would have reached 34 cm only if d had been 0.

EXAMPLE 8

Two parallel chords of a circle of radius 10 cm are 16 cm and 12 cm long. How far apart are they?

Take each chord's distance from the centre first. The combining comes second, and it is where the care is needed, because the question as asked has two answers.

the 16 cm chord $d_1 = \sqrt{10^2 - 8^2} = \sqrt{36}$ **6 cm**

the 12 cm chord $d_2 = \sqrt{10^2 - 6^2} = \sqrt{64}$ **8 cm**

Nothing in the question says which side of the centre each chord is on, and the two arrangements do not agree:

opposite sides $6 + 8$ **14 cm**

the same side $8 - 6$ **2 cm**

Both are right. A question like this is not finished until you say which case you are in — or give both.

It settles the comparison of unequal chords too, which no amount of congruence was going to give us.

THEOREM 8

Of two chords of a circle that are not equal, the longer lies nearer the centre.

Suppose $AB > DE$, at distances d_1 and d_2 from the centre. Then $2\sqrt{r^2 - d_1^2} > 2\sqrt{r^2 - d_2^2}$, so $r^2 - d_1^2 > r^2 - d_2^2$, so $d_1^2 < d_2^2$ — and since distances are never negative, $d_1 < d_2$. The longer chord is the nearer one.

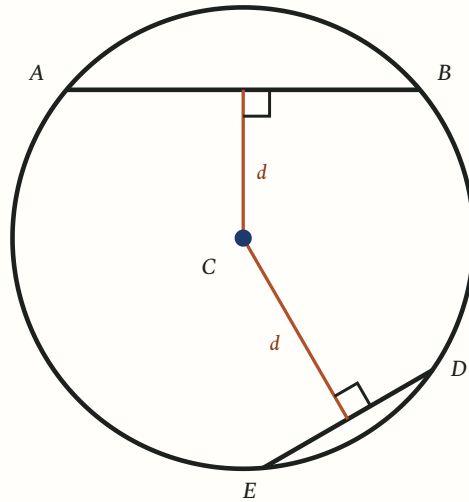


Fig. 5.9 The longer chord AB lies closer to the centre than the shorter chord DE .

The two extreme cases are worth following to the end. Slide a chord towards the centre until $d = 0$: the formula gives length $2r$, and the chord has become a diameter. That is Section 5.2's result — no chord is longer than a diameter — arriving by an entirely different road. Slide it the other way until $d = r$ and the length falls to zero: the chord has shrunk to a single point of the circle. Which is the answer to the question left hanging in Section 5.2. There is no shortest chord, because the moment a chord is short enough to be a candidate it has stopped being a chord at all.

EXAMPLE 9

A circle has radius 25 cm. One of its chords is 48 cm long and another is 14 cm long. Which lies nearer the centre, and by how much?

Theorem 8 answers the first half before any arithmetic starts: the longer chord is the nearer, so the 48 cm chord is the closer of the two. Only the *how much* needs calculating, and for that we halve each chord and use the right-angled triangle again.

the 48 cm chord $\sqrt{25^2 - 24^2} = \sqrt{49}$ **7 cm**

the 14 cm chord $\sqrt{25^2 - 7^2} = \sqrt{576}$ **24 cm**

The longer chord sits 7 cm out and the shorter 24 cm, a difference of 17 cm — the ordering Theorem 8 promised before any arithmetic began.

EXERCISE SET 5.5

1. Fill in the blanks. In a circle of radius 13 cm, a chord lying 5 cm from the centre is _____ cm long; and a chord of that circle 10 cm long lies _____ cm from the centre.
2. Find the length of a chord of a circle of radius 7 cm that lies 6 cm from the centre.
3. Explain, using Fig. 5.8, why a chord lying d from the centre of a circle of radius r has length $2\sqrt{r^2 - d^2}$.
4. One chord of a circle of radius 10 cm lies 6 cm from the centre, another lies 8 cm from it. Which is longer, and by how much?
- *5. In a circle, the chord AB is twice as far from the centre as the chord CD . Does it follow that $CD = 2AB$? Give reasons, and a counter-example if there is one.
- *6. What happens to the length of a chord as its distance from the centre creeps up towards r ? And what is the greatest number of chords of a given length a circle can have?

5.8

ANGLES SUBTENDED BY AN ARC

Everything so far has been about chords. But a chord comes with arcs attached — two of them — and an arc subtends angles of its own.

An **arc** is a connected piece of a circle, fixed by its two end points. Two points A and B cut a circle into two arcs, and unless AB is a diameter one is shorter than the other: the shorter is the **minor arc**, the longer the **major arc**. Naming a third point says which you mean — in Fig. 5.10, arc AXB is the minor one and arc AYB the major.

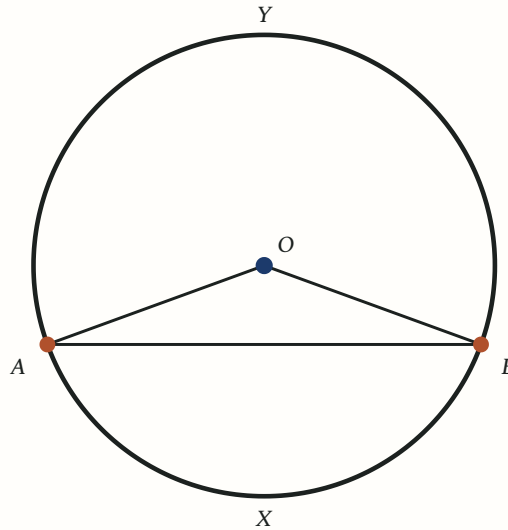


Fig. 5.10 The chord AB cuts the circle into the minor arc AXB and the major arc AYB .

The angle an arc subtends at the centre is the angle swept out as you travel along it: start with the radius at A , move round to B the way the arc goes, and measure how far the radius has turned. For the minor arc that is the ordinary angle $\angle AOB$. For the major arc you sweep the other way and get the reflex angle instead. The two together make a complete turn.

Now a question that is not about the centre at all. Take the arc AXB , and a point D lying on the circle but *not* on that arc. Join D to A and to B . How big is $\angle ADB$?

You would expect the answer to depend on where D sits. It does not. Slide D anywhere along the rest of the circle and the angle under which it sees AB does not move at all. Check it before reading on — draw a circle, fix a chord, and measure the angle at three or four points of the far arc.

THEOREM 9

The angle an arc subtends at the centre of a circle is twice the angle it subtends at any point of the circle outside the arc.

Let the arc be AFB with centre C , and let D be a point of the circle outside it. Join D to C and carry the line on until it meets the circle again, at E . Two isosceles triangles now do all the work, because CA , CB and CD are all radii.

In $\triangle BCD$: $CB = CD$, so $\angle CBD = \angle CDB$.

$\angle BCE$ is the exterior angle of $\triangle BCD$ at C , so $\angle BCE = \angle CBD + \angle CDB = 2\angle BDC$.

In $\triangle ACD$: $CA = CD$, so $\angle CAD = \angle CDA$, and in the same way $\angle ACE = 2\angle ADC$.

Adding the two results, $\angle BCA = \angle BCE + \angle ECA = 2\angle BDC + 2\angle ADC = 2\angle BDA$, which is the theorem.

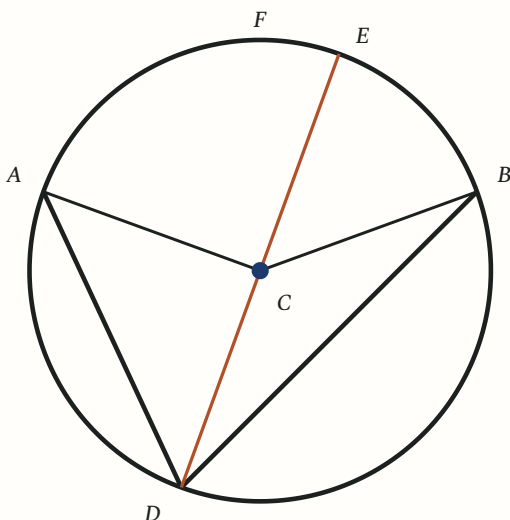


Fig. 5.11 The arc AFB subtends $\angle ACB$ at the centre and $\angle ADB$ at D . The first is twice the second.

One step there was taken without comment: that the line DC , carried on, lands on the arc AFB — which is what put E between A and B and let the two angles at C be added. For some positions of D it lands outside the arc instead, and the figure looks different.

Nothing else changes. The same two isosceles triangles appear, and the same exterior-angle step applies to each; only now the angles at C must be subtracted rather than added, and so must the angles at D . That gives $\angle ACB = \angle ACE - \angle BCE = 2\angle ADC - 2\angle BDC = 2\angle ADB$, and the theorem stands in that case too. It is worth drawing.

Theorem 9 is stated about the centre, but the useful part is what it says about everywhere else. The angle at the centre plainly does not depend on D — the

centre does not know where D is. So the angle at D , being exactly half of it, cannot depend on D either.

ANGLES IN THE SAME SEGMENT

All the points of a circle lying outside a given arc see that arc's chord under the same angle.

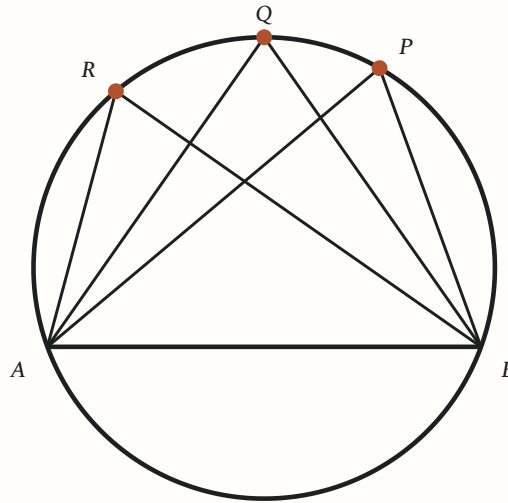


Fig. 5.12 $\angle APB$, $\angle AQB$ and $\angle ARB$ are equal — and so is the angle at every other point of the major arc.

EXAMPLE 10

An arc AXB subtends 110° at the centre of a circle. What angle does it subtend at a point of the major arc? And what angle does the *other* arc subtend at a point of the minor arc?

The first is Theorem 9 read straight off. The second asks about a different arc, so the angle to halve is the one that arc sweeps at the centre — the reflex one.

arc AXB 110° at the centre, halved 55°

the other arc $360^\circ - 110^\circ = 250^\circ$, halved 125°

A point on the major arc sees the chord at 55° ; a point on the minor arc sees it at 125° . Notice that the two add to 180° . That is not a coincidence of these numbers — the two central angles always make a full turn, so their halves always make a straight angle. Section 5.9 turns the observation into a theorem.

One case of Theorem 9 is used so often that it has a name of its own. A **corollary** is a fact that drops out of a result already proved, with no fresh argument needed.

COROLLARY

The angle subtended by a diameter at any point of the circle is a right angle.

Let AB be a diameter and D any other point of the circle. The arc from A to B that avoids D is a semicircle, and the angle it sweeps at the centre is a straight angle, 180° . By Theorem 9 the angle at D is half of that: $\angle ADB = 90^\circ$, wherever on the circle D happens to be.

Section 5.4 left a debt here. It claimed that the circumcentre of a right-angled triangle lands on the midpoint of the hypotenuse, and gave no reason. The corollary is half of the reason. The other half — that a point seeing AB under a right angle has no choice but to lie on the circle with AB as diameter — is exactly the sort of statement Section 5.9 is about, and you will be able to finish the argument there.

EXAMPLE 11

AB is a diameter of a circle and C is a point on the circle, with $\angle ABC = 35^\circ$. Find $\angle BAC$.

Nothing is said about the angle at C — and nothing needs to be, because AB is a diameter, and the corollary fixes that angle wherever on the circle C happens to sit.

angle at C	the angle in a semicircle	90°
angle at B	given	35°

The three angles of $\triangle ABC$ add to 180° , so

$$\angle BAC = 180^\circ - 90^\circ - 35^\circ = 55^\circ$$

The corollary did all the work: without it, one angle and a diameter would not determine the triangle at all.

EXERCISE SET 5.6

1. In a circle of radius 12 cm with centre O , the angle $\angle AOB$ is 60° . How long is the chord AB ?
2. An arc subtends 70° at the centre. What angle does it subtend at a point of the circle outside the arc?
3. An arc subtends 250° at the centre, measured the long way round. What angle does it subtend at a point of the circle outside it?
4. Let A and B be points of a circle.
 - (i) Are there points X and Y on the same side of AB with $\angle AXB$ different from $\angle AYB$?
 - (ii) If $\angle AXB = \angle AYB$, must X and Y be on the same side of AB ?
- *5. A chord of a circle is exactly as long as the radius. What angle does it subtend at the centre, and what angle at a point of the major arc?

5.9 POINTS ON ONE CIRCLE

Section 5.4 counted the circles through two points — infinitely many — and through three, where there is exactly one so long as they are not in a row. Four points are a different matter altogether. Any three of them already fix a circle, so the fourth has no freedom left: either it happens to land on that circle or it does not. Points that all lie on one circle are called **concylic**.

So when does the fourth point land? Theorem 9 supplies the clue. If A , B , C and D are concyclic, with C and D on the same side of AB , then C and D are on the

same arc, and angles in the same segment are equal: $\angle ACB = \angle ADB$. That much is forced. The question is whether it works in reverse.

THEOREM 10

If a segment AB subtends equal angles at two points C and D lying on the same side of it, then A, B, C and D are concyclic.

The proof takes a shape not used anywhere else in this chapter: rather than building the circle, we show that D has nowhere else to be.

The points A, B, C are not collinear — if they were, $\angle ACB$ would be a straight angle or nothing at all — so by Theorem 1 there is exactly one circle through them. Draw it. Now D is either outside that circle, or inside it, or on it. Rule out the first two and only the third is left.

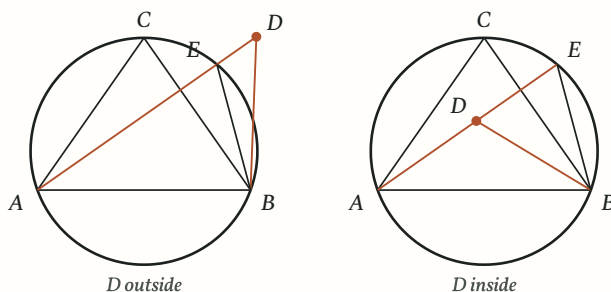


Fig. 5.13 Neither case can happen. Each forces $\angle ACB$ to be bigger, or smaller, than itself.

Suppose D lies outside. Then the segment AD crosses the circle at some point E . Now C and E are both on the circle and on the same side of AB , so $\angle ACB = \angle AEB$. But $\angle AEB$ is an exterior angle of $\triangle BED$, and an exterior angle is larger than either interior angle opposite it, so $\angle AEB > \angle ADB$. Putting the two together, $\angle ACB > \angle ADB$ — and we were given that those two angles are equal. So $\angle ACB$ would be greater than itself.

Suppose instead D lies inside. Extend AD until it meets the circle at E . This time $\angle ADB$ is the exterior angle of $\triangle BED$, so $\angle ADB > \angle AEB = \angle ACB$, and the same impossibility arrives from the other direction.

Both positions are ruled out, so D lies on the circle through A , B and C . The four points are concyclic.

A quadrilateral whose four vertices are concyclic is called a **cyclic quadrilateral**. Theorem 10 tells us when four points form one; the next result tells us what we get in return.

THEOREM 11

The two opposite angles of a cyclic quadrilateral add up to 180° .

Let $ABCD$ be cyclic, with centre O , and consider the two opposite angles at A and at C . Each is an angle subtended by an arc at a point of the circle outside it, so Theorem 9 applies to both — and the trick is to be careful about *which* arc.

$\angle BAD$ is subtended at A by the arc from B to D that passes through C . So it is half the angle that arc sweeps at O .

$\angle BCD$ is subtended at C by the arc from B to D that passes through A . So it is half the angle *that* arc sweeps at O .

Between them the two arcs make up the whole circle, so the two angles they sweep at O add to a complete turn of 360° . Halving,

$$\angle BAD + \angle BCD = \frac{1}{2} \times 360^\circ = 180^\circ$$

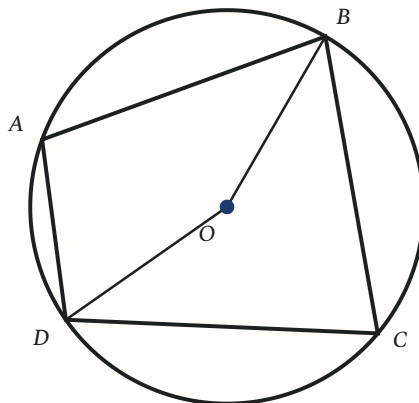


Fig. 5.14 A cyclic quadrilateral. The arcs BAD and BCD together make the whole circle.

The same holds for the other pair, $\angle ABC$ and $\angle ADC$, by the identical argument on the arcs through D and through B . And the converse is true as well.

EXAMPLE 12

A cyclic quadrilateral $ABCD$ has $\angle A = 95^\circ$ and $\angle B = 70^\circ$. Find its other two angles.

Theorem 11 pairs each angle with the one *opposite* it, never with a neighbour. So A goes with C , and B with D .

$$A \text{ with } C \quad \angle C = 180^\circ - 95^\circ \quad 85^\circ$$

$$B \text{ with } D \quad \angle D = 180^\circ - 70^\circ \quad 110^\circ$$

Check the total: $95 + 70 + 85 + 110 = 360$, which the angles of any quadrilateral must come to. That check is worth doing every time, because it catches the commonest slip here — pairing angles that sit next to each other instead of across from each other.

THEOREM 12

If two opposite angles of a quadrilateral add up to 180° , then its four vertices are concyclic.

The proof runs like Theorem 10's — by showing the fourth vertex has nowhere else to go. Suppose $\angle BAD + \angle BCD = 180^\circ$. Since A, B, D are not collinear there is a circle through them; suppose C lay outside it, and let E be where DC meets the circle. Then $ABED$ is cyclic, so Theorem 11 gives $\angle BAD + \angle BED = 180^\circ$. Comparing with what we were given, $\angle BED = \angle BCD$. But $\angle BED$ is an exterior angle of $\triangle BEC$ and so exceeds $\angle BCE$, which is $\angle BCD$. Once again a quantity is greater than itself. The case with C inside is the same argument with the roles exchanged, and is worth writing out.

CLOSING AN OLD DEBT

Twice now this chapter has promised that the circumcentre of a right-angled triangle sits at the midpoint of the hypotenuse. Theorem 10 pays up.

Let $\triangle ABD$ have its right angle at D , and let M be the midpoint of the hypotenuse AB . Draw the circle with centre M and radius MA , so that AB is a diameter of it, and mark any point D' of that circle on the same side of AB as D . By the corollary in Section 5.8, $\angle AD'B = 90^\circ$ — which is also $\angle ADB$. Two points, equal angles, same side: by Theorem 10 the four points A, B, D' and D are concyclic, so D is on that circle too. Its centre M is therefore equidistant from A, B and D — which is exactly what it means to be the circumcentre.

EXAMPLE 13

A quadrilateral has angles $85^\circ, 100^\circ, 95^\circ$ and 80° , taken in order round the shape. Can a circle be drawn through all four vertices?

Theorem 12 is the test, and it is a test on opposite pairs — so the angles have to be paired first and second with third and fourth, not as they are listed.

<i>first and third</i>	$85^\circ + 95^\circ$	180°
<i>second and fourth</i>	$100^\circ + 80^\circ$	180°

Both pairs come to 180° , so by Theorem 12 the four vertices are concyclic and the circle can be drawn. In fact only one pair ever needs checking: all four angles total 360° , so if one opposite pair makes 180° the other pair has no choice but to do the same.

EXERCISE SET 5.7

1. A cyclic quadrilateral has $\angle A = 75^\circ$ and $\angle B = 110^\circ$. Find $\angle C$ and $\angle D$.
2. A quadrilateral has angles $80^\circ, 110^\circ, 100^\circ$ and 70° , in that order round the shape. Can it be cyclic? Can it be drawn at all?
3. $PQRS$ is cyclic with $\angle P = (2x + 10)^\circ$ and $\angle R = (3x - 20)^\circ$. Find x , and then both angles.

- *4. Show that a rectangle is the only parallelogram that can be inscribed in a circle.
- *5. Let $ABCD$ be cyclic and let E lie on CD extended. Explain why the exterior angle $\angle BCE$ equals the interior opposite angle $\angle BAD$.

END-OF-CHAPTER EXERCISES

1. A chord of a circle lies 5 cm from the centre, and the radius is 13 cm. How long is the chord?
2. A circle has diameter 26 cm, and a chord of length 24 cm is drawn in it. How far is that chord from the centre?
3. A chord of length 16 cm lies 6 cm from the centre. Find the radius.
4. Prove that the perpendicular bisector of a chord passes through the centre of the circle.
5. AB is a diameter of a circle and C is another point on it. What is $\angle ACB$? Give the reasoning, not just the number.
6. A cyclic quadrilateral has $\angle A = 75^\circ$ and $\angle B = 110^\circ$. Find $\angle C$ and $\angle D$.
7. A cyclic quadrilateral has sides 5, 5, 12, 12 units, in that order round the shape. Find its area.
8. Two parallel chords, of lengths 10 cm and 24 cm, lie on the same side of the centre and 7 cm apart. Find the radius of the circle.
- *9. Draw a circle in which a chord of length 6 cm stands 3 cm from the centre. What must the radius be?
- *10. Justify the claim that no chord of a circle is longer than a diameter. Give two arguments, one of them from Section 5.7.
- *11. Take all the chords of a circle that have one fixed length. What shape do their midpoints trace out?
- *12. Two chords AB and AC of a circle have the same length. Explain why the centre must lie on the bisector of $\angle BAC$.

- *13.** Let A be a point inside a circle with centre O . Show that the shortest chord through A is the one perpendicular to OA .
- *14.** A regular hexagon is inscribed in a circle of radius r . Find the length of a side, and the distance of each side from the centre.

The starred questions are harder. Most can be settled by picking the right theorem rather than by calculating — and deciding which one is part of the work.

One chapter, twelve theorems, and a single sentence underneath all of them: a circle is the set of points at a fixed distance from a fixed point. Nothing else was ever assumed. Every symmetry, every chord, every angle in this chapter was squeezed out of that one sentence — which is what a good definition is for.

WHAT THIS CHAPTER ESTABLISHED

1. A circle is the locus of points of a plane lying at a fixed distance, the radius, from a fixed point, the centre.
2. A circle has complete rotational symmetry about its centre, and reflection symmetry in every diameter — and in no other line.
3. Infinitely many circles pass through two given points, with their centres strung along the perpendicular bisector of the segment joining them. Through three points that are not collinear there passes exactly one.
4. The segment from the centre to the midpoint of a chord is perpendicular to it, and the perpendicular from the centre bisects it. Of the three facts — from the centre, through the midpoint, at a right angle — any two force the third.
5. Equal chords subtend equal angles at the centre and lie at equal distances from it, and each of those statements holds in reverse. A chord at distance d has length $2\sqrt{r^2 - d^2}$, which settles all of it at once, including that the longer of two chords is the nearer.
6. An arc subtends at the centre twice the angle it subtends at any point of the circle outside it. So every point of the far arc sees a chord under the same angle, and a diameter is seen at a right angle from everywhere on the circle.
7. Four points are concyclic exactly when a segment joining two of them subtends equal angles at the other two on the same side — equivalently, when opposite angles of the quadrilateral they form add to 180° .
8. Measuring confirms; only an argument explains. More than once in this chapter the measurements were right and still proved nothing at all.

THE RESULTS, IN ONE PLACE

1. There is exactly one circle through three non-collinear points.
 2. Equal chords subtend equal angles at the centre.
 3. Chords subtending equal angles at the centre are equal.
 4. The centre joined to the midpoint of a chord is perpendicular to it.
 5. The perpendicular from the centre to a chord bisects it.
 6. Chords of equal length are equidistant from the centre.
 7. Chords equidistant from the centre are of equal length.
 8. Of two unequal chords, the longer lies nearer the centre.
 9. An arc subtends at the centre twice the angle it subtends at a point of the circle outside the arc.
 10. If a segment subtends equal angles at two points on the same side of it, the four points are concyclic.
 11. Opposite angles of a cyclic quadrilateral add to 180° .
 12. If opposite angles of a quadrilateral add to 180° , it is cyclic.
-



Next year the circle gives up more: tangents, the lengths cut off when two chords cross, and circles that touch one another. All of it comes from the same one sentence.
